

Comprehensive Characterization of Oxidative Stress-Modulating Chemicals Using GPT-Based Text Mining

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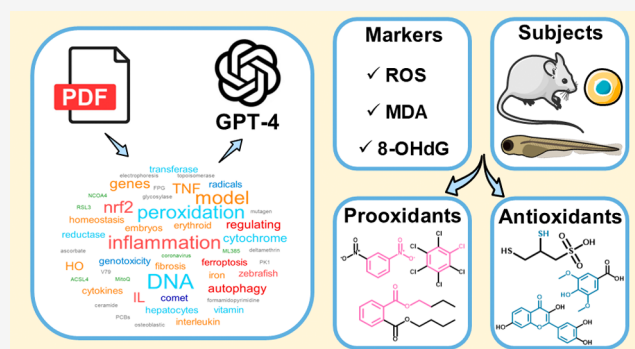
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ABSTRACT: The screening of hazardous environmental pollutants is hindered by the limited availability of toxicological databases. Large language model (LLM)-based text mining holds the potential to automatically extract complex toxicological information from the literature. Due to its relevance to diseases and the challenge of comprehensive characterization, oxidative stress serves as a suitable case for research by text mining. In this study, a robust workflow utilizing a LLM (i.e., GPT-4) was developed to extract information on oxidative stress tests, including data collection, text preprocessing, prompt engineering, and performance evaluation procedures. A total of 17,780 relevant records were extracted from 7166 articles, covering 2558 unique compounds. A rising interest in oxidative stress was observed over the past two decades. A list of known prooxidants ($n = 1416$) and antioxidants ($n = 1102$) was established, with the leading chemical categories being pharmaceuticals, pesticides, and metals for prooxidants and pharmaceuticals and flavonoids for antioxidants. Structural alert analysis identified potential prooxidant (e.g., chlorobenzene, nitrobenzene, and tertiary amines) and antioxidant (e.g., flavonoid and thiol) substructures. These findings illustrate the feasibility of building toxicological databases through LLM-based text mining in a cost-efficient manner, and the information obtained from the technique holds significant promise for future applications in environmental and health research.

KEYWORDS: artificial intelligence, database, reactive oxygen species, malondialdehyde, 8-hydroxy-2'-deoxyguanosine, prooxidants, antioxidants, structural alerts



1. INTRODUCTION

As an advanced strategy for exploring the relationship between environmental pollution and human health, identification and prioritization of hazardous pollutants using suspect and nontarget screening have received substantial attention. In this field, the databases compiling chemicals targeting specific toxicological end points play a crucial role by directly providing suspect lists and facilitating hazardous risk assessment.^{1,2} These databases can be extended by incorporating multiple technologies, such as predicting the effects of nonstudied chemicals through machine-learning models or structural alerts (SAs) and forecasting adverse outcomes based on the databases of relevant molecular initiating events (MIEs) and key events (KEs).^{3–5} Currently, there are limited resources of publicly available toxicological databases. One of the primary resources is open data from the interdisciplinary community that conducts in vitro high-throughput screening tests, including Toxicology in the 21st Century (Tox21),⁶ the Broad Institute,⁷ and the Scripps Research Molecular Screening Center.⁸ Another resource is literature information from manually curated databases such as ChEMBL,⁹ which is

heavily constrained by high labor and time costs. Automatic extraction of literature content using text mining tools provides a comprehensive and easy access to the experimental results accumulated by numerous researchers over time and is thus a flexible and promising approach to provide valid data for multiple unattended toxicological effects or compounds of interest.

Traditionally, text mining has focused on pattern-/rule-based methods and supervised learning methods, both of which exhibit limitations in detecting relations among entities, relying heavily on manual encoding of complex rules and manual annotation of training corpora, respectively. Prior attempts to extract chemical–biomedical entity relations have mostly been limited to binary or ternary relations and have

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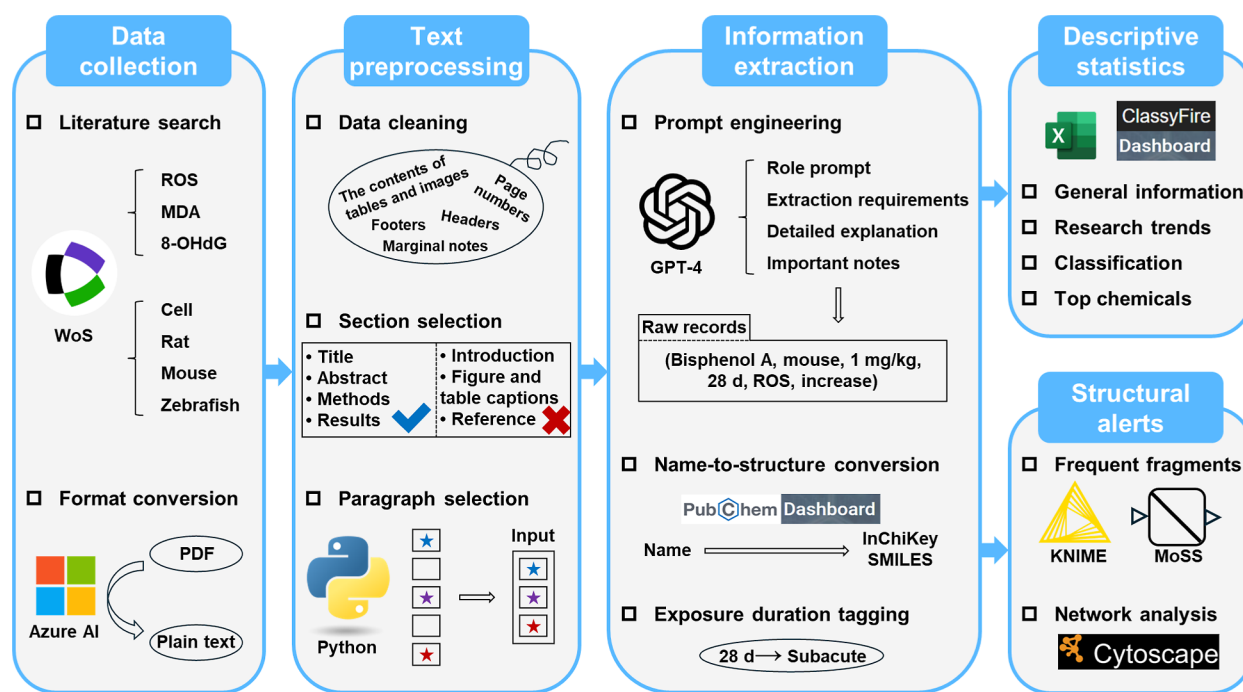


Figure 1. LLM-based workflow mainly consists of data collection, text preprocessing, information extraction, descriptive statistics, and SA analysis. “WoS” stands for Web of Science.

rarely achieved precision and recall of 80%.¹⁰ In comparison, the emergence of large language models (LLMs) has recently revolutionized text mining technology by directly extracting semantic information.¹¹ The generative pretrained transformer (GPT) is an advanced LLM capable of understanding and generating human-like text, particularly suitable for interactive textual tasks.¹² Recently, it was used for automated text mining in the area of materials chemistry to establish data sets containing multiple entities (i.e., the parameters of metal–organic framework synthesis), achieving impressively high precision and recall (>90%).^{13,14} Although promising, the application of LLMs for obtaining toxicological information has not yet been explored.

Oxidative stress implies an imbalance between oxidants and antioxidant defenses in the body, which may cause biological damage. It is involved in numerous MIEs and KEs across various toxicological pathways and is intimately associated with many diseases including atherosclerosis, chronic obstructive pulmonary disease, Alzheimer’s disease, and cancer.¹⁵ The chemical modulators of oxidative stress have become an interest for research with abundant publications. On the one hand, a variety of environmental pollutants can elevate levels of reactive oxygen species (ROS), leading to oxidative damage to proteins, lipids, and deoxyribonucleic acid (DNA), ultimately triggering inflammation and cell death.^{16–18} On the other hand, several antioxidants such as ROS scavengers exhibited therapeutic potential for oxidative damage in preclinical studies.¹⁵ However, there is currently a lack of an integrated list of oxidative stress-modulating chemicals that could aid in the screening of environmental pollutants and prioritization of compounds with significant hazards in environmental and health research.

Thus, a refined LLM-based workflow targeting oxidative stress-modulating chemicals was established and evaluated. Data collection, text preprocessing, and prompt engineering were conducted to interface and guide GPT-4, the most

advanced version with an application programming interface (API) as of June 2024, to extract information on chemicals, markers, subjects, exposure conditions, and results of oxidative stress tests. The objectives of this study were (1) to establish a list of known prooxidants and antioxidants, (2) to explore the current trends of oxidative stress-related research, and (3) to identify substructures potentially related to the oxidative stress effects.

2. MATERIALS AND METHODS

2.1. Data Collection. A literature search was conducted through Web of Science on September 27, 2023, specifically utilizing the Core Collection database with the SCI-EXPANDED edition (Figure 1). The search covered the categories of “Environmental Sciences”, “Chemistry, Medicinal”, and “Toxicology” to ensure a thorough exploration. The document type was limited to articles, and the language was limited to English.

A query (Appendix S1) was designed to search for articles related to specific entities of oxidative stress markers and experimental subjects. Guided by the criteria of being able to reliably differentiate between prooxidants and antioxidants, three of the most widely used markers that reflect oxidative conditions were selected, including ROS, malondialdehyde (MDA), and 8-hydroxy-2'-deoxyguanosine (8-OHdG), while those stress-responsive indicators such as antioxidant enzyme activities and endogenous nonprotein antioxidant levels were excluded from further investigation.^{19–21} Similarly, based on physiological relevance to humans and broad utility in toxicological research, cells, rats, mice, and zebrafish were chosen as experimental subjects.^{22–24} Concentrating on the research results of these essential markers and subjects helps to reduce the complexity of text mining tasks.

Additionally, population studies were excluded because of a tendency to focus on statistical correlations rather than on direct experimental evidence. Studies pertaining to nano-

particles and microplastics were also excluded to allow for a more focused and comprehensive characterization of small-molecule compounds. It was found that only a small number of related studies were published between 1976 and 1999 (approximately 3% of all the literature obtained from the query search), and a notable upward trend was observed since 2000. Thus, a time scope encompassing 2000–2023 was chosen for further discussion. The basic information on articles, including the DOI, publication year, title, abstract, and keywords, was batch-exported, and the available portable document format (PDF) documents were downloaded.

2.2. Text Mining. **2.2.1. File Format Conversion and Text Preprocessing.** To interface with the API of GPT-4, PDF documents were converted into plain text in advance using a commercial parser (Azure AI, Microsoft, Redmond, WA), which applies optical character recognition to reliably extract text blocks (i.e., paragraphs) in a layout-aware manner. A text preprocessing procedure consisting of three major steps was implemented in Python to reduce the cost of using GPT-4 and improve information extraction efficiency by eliminating irrelevant content.

First, data cleaning was conducted. The issue with the commercial parser is that contents from tables and images, headers, footers, marginal notes, and page numbers are often recognized as paragraphs, even if the contents are very short. To eliminate such information embedded in the main text, any paragraph with 10 words or fewer was deleted. Headings were identified and preserved by the presence of predefined named entities such as “introduction”, “method”, “result”, and “reference”.

Second, section selection was carried out. Based on the common structure of articles, the required information is usually concentrated in the Title, Abstract, **Materials and Methods**, and **Results and Discussion** sections. Hence, the additional sections such as figure and table captions, Introduction, and References were distinguished and removed.

Finally, paragraph selection was applied to the **Materials and Methods** and **Results and Discussion** sections to further enhance the information density. Paragraphs rather than sentences were taken as the minimal unit to ensure semantic integrity and prevent information loss or misinterpretation. Whether a paragraph was selected depended on the presence and position of predefined named entities representing four elements: chemical, marker, subject, and exposure condition (i.e., method, duration, and dose). For instance, a paragraph would be selected if any marker-related named entity (e.g., “ROS”, “8-OHdG”, and “MDA”) appeared. Similarly, if a chemical-related named entity (e.g., “chemical” and “reagent”) appeared within the first five words of a paragraph, the word count would be considered. Paragraphs with more than ten words were selected, while shorter paragraphs were deemed subheadings, prompting the selection of the next paragraph.²⁵ The selected paragraphs were then concatenated in their original order to form the input text. All the codes for PDF parsing, text preprocessing, and API running can be assessed at https://github.com/liangwenqing1996/GPT-4_based_text_mining.

2.2.2. Prompt Engineering. Prompt engineering has become an essential technique for optimizing the utility and accuracy of LLMs.²⁶ A well-constructed prompt with clear logic, detailed rules, and repeated reminders of common mistakes was used to guide the responses of GPT-4, especially to counteract the challenges of machine hallucinations. Role-

prompting is an important part in prompt engineering.²⁶ It was used to guide GPT-4 to focus on oxidative stress-related information in environmental science, toxicology, and pharmacology through specific instructions ([Appendix S2](#)).

The body of the prompt consisted of three major parts: extraction requirements, detailed explanation, and important notes. A structured output format was specified in the first part, which defined each output as a record containing information relevant to oxidative stress tests including the elements of chemicals, markers, subjects, exposure conditions, and results. Further, the second part provided precise rules for determining each output element using the following techniques: (1) giving thorough descriptions of situations to be included and excluded, (2) clearly enumerating the logical connections between the current element and other elements, (3) presetting fixed output values or possible output value references or specific format requirements for each element, (4) offering common examples for the involved terminology, and (5) providing explanations for important abbreviations and units related to the task. The third part was designed to additionally emphasize possible mistakes, which was quite effective in improving the performance of GPT-4. For example, by simply adding the rule not to make up any entries, the number of instances of machine hallucinations was largely reduced. Full text of the prompt is shown in [Appendix S2](#).

2.2.3. Algorithm Performance Evaluation. To verify the algorithms of text preprocessing and information extraction using GPT-4, 35 pieces of literature containing required information and another 35 pieces not containing the information were randomly selected to serve as a validation set. The number of documents was determined using Cochran's formula, which is a statistical method widely used to ensure the reliability of sampling size from a large population.^{27,28} A margin of error of 10% with a confidence level of 95% was applied, and detailed calculations can be found in [Appendix S3](#).

For documents containing required information, the precision, recall, and F1 score were calculated via [eqs 1–3](#):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (3)$$

where TP, FP, TN, and FN represent true positives, false positives, true negatives, and false negatives, respectively. Precision is the proportion of correct results among all of the returned results. Recall is the ratio of correctly retrieved relevant instances to the total number of relevant instances. The F1 score integrates precision and recall, with a higher value indicating a better performance of the workflow. For documents not containing required information, the frequency of correctly returning “No results” was calculated.

2.2.4. Data Postprocessing and Filtering. The extracted records were processed through fact and logic checking to exclude fictitious elements and unreasonable content. Specifically, records with missing or ambiguous chemical names were excluded. Records lacking critical information (i.e., subject, marker, and result) or those that failed to return the specified output values were also removed. Detailed subject information

and exposure conditions were manually reviewed, and any unreadable content (e.g., exposure doses or times lacking units) was excised, although the corresponding records were retained. Further, a name-to-structure conversion procedure was conducted to map various chemical entity aliases back to a single structure. The CompTox Chemicals Dashboard²⁹ and PubChem³⁰ were utilized to verify InChiKey and SMILES, and records with unverifiable structures were discarded. Additionally, exposure duration was tagged as acute (within 24 h), subacute (1 day to 1 month), subchronic (1 to 3 months), and chronic (3 months to 2 years).

3.2.3. Structural Alert Analysis. The SAs are substructures having potential link with specific effects, characterized by a higher frequency in active compounds than in inactive analogues.³¹ A procedure built on the KNIME Analytics Platform v.5.2.3 (KNIME, Zurich, CH) allowed for the finding of highly frequent fragments in small molecules using the MoSS node (Figure S1 and Appendix S4), which was used for exploring direct mechanistic relationships with oxidative stress through expert judgment.³² The input consisted of SMILES and binary activity categories, and the output content included SMILES of frequent fragments with corresponding counts. Moreover, network analysis for the co-occurrence of the obtained SAs was conducted using Cytoscape v3.10.2.³³

3. RESULTS AND DISCUSSION

3.1. Performance of the Workflow. When prompt engineering is sufficiently structured, remarkable results in complex text mining tasks involving multiple entities and network interactions can be achieved. For documents containing required information in the evaluation set, the precision, recall, and F1 score were 0.41, 0.91, and 0.57 during the text preprocessing procedure, respectively. The objective in this procedure was to appropriately reduce the text length while retaining the essential information. Thus, high recall with moderate precision was emphasized. In contrast, during the information extraction procedure, the correctness of the results was prioritized over comprehensiveness. Therefore, while both high precision and high recall would be desired, a slightly lower recall was acceptable. Through optimization of prompt engineering, the values of 0.91, 0.81, and 0.86 were finally achieved for precision, recall, and F1 score, respectively. Moreover, the frequency of correctly returning “No results” was 97% for documents not containing the required information, indicating satisfactory performance in avoiding false positives.

A retrospective analysis was also conducted to investigate the causes of false negatives and false positives. For documents containing required information, false negatives primarily occurred when the information in the textual input was incomplete or the semantic content was presented in complex structures, such as overly fragmented information or implicit statements. False positives were mainly caused by the extraction of records containing unspecified subjects or markers, which can be addressed through postprocessing and filtering procedures. Notably, a small subset of these false positives (2.5% of all extracted records) was attributed to machine hallucinations. These hallucinations tended to occur in cases of irregular experimental designs, where chemicals, subjects, and markers were not consistently matched one-to-one. Then, the GTP model was inclined to fill in gaps. For instance, one study measured ROS changes induced by five

compounds, as well as 8-OHdG changes by three of those compounds.³⁴ Although absent in the input text, the hallucinatory information on the records of 8-OHdG changes for the other two compounds was returned. Differently, for documents not containing the required information, false positives were exclusively due to the extraction of records containing unspecified subjects or markers, with no instances of hallucinations observed. Overall, despite the challenges posed by complex semantic structures and irregular experimental designs, the performance of the workflow met the expectations for reliably extracting information relevant to oxidative stress tests with high precision and minimal hallucinations.

3.2. General Information of the Text Mining Results.

A total of 17,780 relevant records were retrieved across 7166 articles by text mining, covering 2558 unique compounds. Among these compounds, the top three by the number of records (i.e., hydrogen peroxide, *N*-acetyl-L-cysteine, and ascorbic acid) were reference chemicals in experimental design with well-known prooxidant or antioxidant properties, which are widely used to induce oxidative damage models or to provide further evidence for the mediation of oxidative stress in adverse outcomes. After excluding information on these reference chemicals, 11,680 records remained (Figure 2).

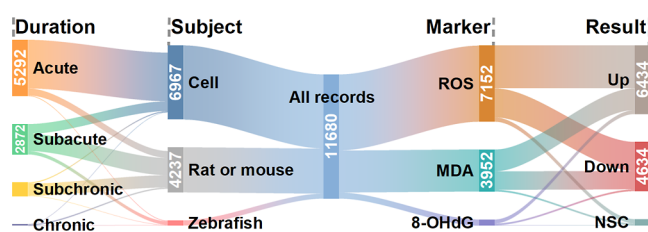


Figure 2. Sankey diagram showing the constitution of the records obtained from text mining. The oxidative stress marker with the highest number of records was ROS (61%), followed by MDA (34%) and 8-OHdG (5%). The proportion of records indicating upregulated oxidative stress (55%) was slightly higher than those indicating downregulated oxidative stress (40%), and few records reported “no significant change” (NSC, 5%). In terms of experimental subjects, cells, rats or mice, and zebrafish accounted for 60%, 36%, and 4%, respectively. Shorter exposure durations had more related records, with acute, subacute, subchronic, and chronic exposures accounting for 45%, 25%, 11%, and 1%, respectively.

These records were organized into an Excel file for open access (Table S1), which provides the DOI of source articles, details on molecular structures and chemical categories, and experimental parameters and results of the oxidative stress tests.

The existing publicly accessible databases for ROS, MDA, and 8-OHdG are provided by ChEMBL and the Broad Institute. These databases mainly serve for drug design, with most of the tested drug-like molecules being novel compounds lacking common names. In the present study, compound names were deemed necessary information to focus on relatively well-known compounds that are usually used in commerce or are found in the environment. Consequently, there was a minimal overlap between the obtained records and the existing drug design databases. To our knowledge, this is the first effort to establish a comprehensive list of ROS, MDA, and 8-OHdG modulators with a focus on environmental compounds.

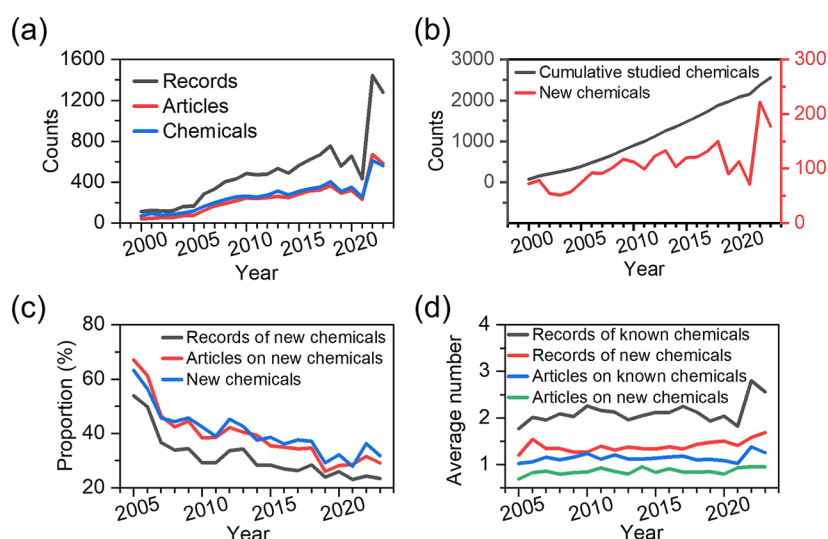


Figure 3. Line charts reflect the research trends. A record refers to an entry obtained from text mining, with each article potentially yielding one or multiple records. (a) The number of records, articles, and chemicals continued to grow annually from 2000 to 2023. (b) The number of new chemicals fluctuated between 50 and 200 annually, and the cumulative number of studied chemicals was steadily increasing from 2000 to 2023. (c) The proportion of new chemicals, related articles, and related records per year continued to decline from 2005 to 2023. (d) The average number of articles and records per known chemical were consistently about 1.5 times higher than that for new chemicals from 2005 to 2023. A “known chemical” refers to a chemical that was already studied before the given year. A “new chemical” refers to a chemical that was studied for the first time in the given year. “Cumulative studied chemicals” refer to all compounds that have been studied up to the given year.

There was a high degree of overlap among the compounds correlated with different markers. For instance, 2,144, 1,285, and 616 compounds were found to be associated with ROS, MDA, and 8-OHdG in separate studies, respectively. Among the compounds associated with ROS, 70% were related to MDA and 86% to 8-OHdG. This phenomenon indicates that changes in the three markers are interrelated, as supported by the prior knowledge that MDA and 8-OHdG are the oxidative damage products of ROS attacking lipids and DNA, respectively.¹⁹

The exposure durations varied across the experimental subjects. Cells (i.e., in vitro assays) were frequently used for acute exposure studies (66%), rodent models (i.e., rat or mouse) were a choice for subacute (38%) and subchronic (27%) exposures, and zebrafish were also selected for subacute (69%) assessments. It is reasonable that the nature of rapid response to chemical stimuli makes the in vitro assays suitable for acute effect evaluation, helping to understand the initial mechanisms of oxidative stress at the cellular level. By contrast, subacute and subchronic experimental designs based on rodent models reflect an interest in the cumulative effects of long-term and repeated exposure, especially in how the modulation of oxidative stress affects physiology, metabolism, and chronic pathological states. In addition, zebrafish are often used in subacute exposure experiments to reveal the impact of oxidative stress modulators on critical developmental stages of embryos. The choices of experimental subjects and the corresponding exposure durations reflect varying research needs regarding the broad biological impacts of oxidative stress modulators.

3.3. Exploration of Research Trends. Using the LLM-based text mining, it is possible to gain insights into the dynamic landscape of oxidative stress-related studies and reveal current progress and potential drawbacks. Research interest in oxidative stress has steadily increased over the past two decades. The number of records, source articles, and studied chemicals was growing in parallel, with a sharp rise in the last

two years (i.e., 2022–2023) (Figure 3a). To identify the drivers of this rise, changes in word frequency within the titles, abstracts, and keywords were analyzed, comparing articles published in 2000–2021 and 2022–2023 (Figure S2). Entities with a more than 2-fold increase in frequency, as well as the emerging ones, indicated rising attention to the inflammation (i.e., “inflammation”, “interleukin”, and “TNF”), the NF-E2-related factor 2/antioxidant response element (Nrf2/ARE) pathway (i.e., “Nrf2”, “heme oxygenase”, “ML385”, and “xCT”), and ferroptosis (i.e., “ferroptosis”, “iron”, “RSL3”, “ACSL4”, and “NCOA4”). The emerging entities also shed light on mitochondrial toxicity (i.e., “MitoQ”), cardiac developmental toxicity (i.e., “GATA4”), and COVID-19 (i.e., “coronavirus”). Overall, expanded and deepened understanding of the pathways, mechanisms, and adverse outcomes associated with oxidative stress is likely the cause of a recent surge in research interest.

Notably, the application of zebrafish exhibited a clear upward trend (Appendix S5 and Figures S3–S7). There were almost no relevant records from 2000 to 2012, a range from 2% to 3% between 2013 and 2017, and a rapid increase from 6% to 10% in 2018–2023. This trend aligns with the development of zebrafish as a model organism for understanding human biology and disease.³⁵ Several key achievements have contributed to this development, including the discovery of 70% vertical homology with human genes in 2013³⁶ and the systematic genome editing of chromosome 1 in 2019, with one-quarter of the mutants related to human diseases.³⁷ Besides, zebrafish have inherent advantages over traditional rodent models (such as lower costs, shorter cycles, higher throughput, and transparent bodies that ease morphological observations), providing another explanation for their increasing popularity.

Another interesting observation was the tendency to repeatedly study the same compounds. The top 100 most focused compounds (4% of the total) accounted for 40% of the records, and the top 500 compounds (20%) accounted for

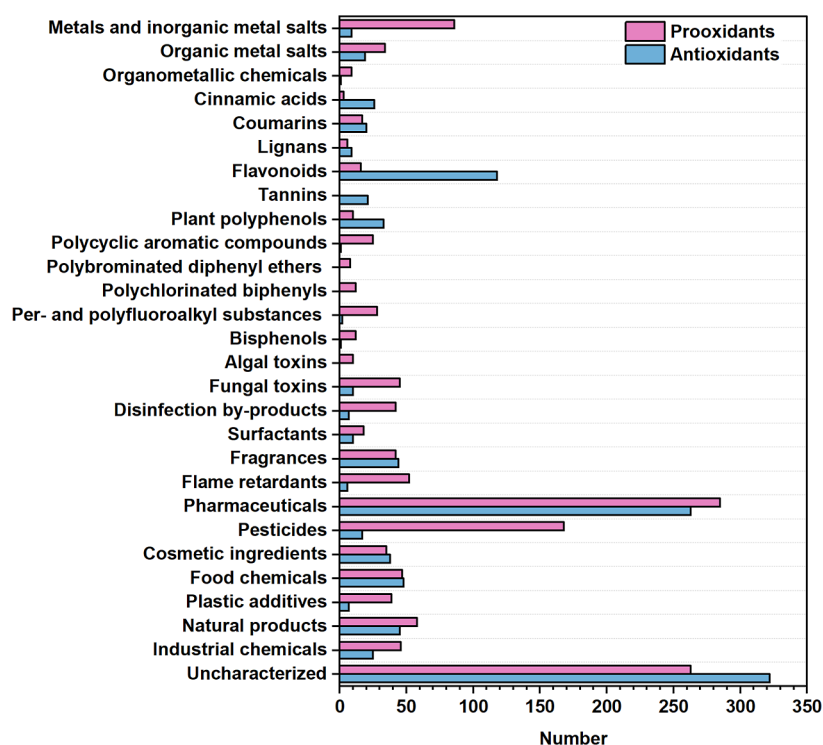


Figure 4. Bar chart showing the number of prooxidants and antioxidants in different chemical categories. List comparison was performed in the order from top to bottom of the y-axis, with each compound being assigned a unique category. If a compound belonged to multiple categories, it was counted only in the foremost one. “Uncharacterized” refers to chemicals that do not appear in any of the referred databases.

70%. This phenomenon reflects the common logic in toxicology that specifically acting compounds receive more attention. Meanwhile, the exploration of less-known chemicals involves inherent uncertainties and extended timelines. The cumulative number of known compounds has steadily increased since 2000 (Figure 3b), with the number of new chemicals (studied for the first time in a given year) fluctuating between 50 and 200 annually. Consequently, when examining the proportion of new chemicals each year (Figure 3c), a synchronized downward trend could be observed, decreasing to around 30% in the past five years (2018–2023). The average number of articles and records per known chemical (already studied before the given year) was consistently higher than that of new chemicals by approximately 1.5 times (Figure 3d). In this case, conducting a comprehensive characterization of known oxidative stress-modulating chemicals would be beneficial, facilitating exploration of new active compounds.

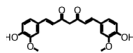
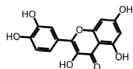
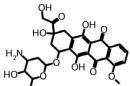
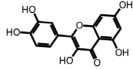
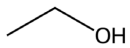
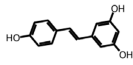
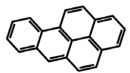
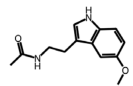
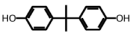
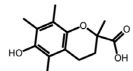
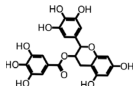
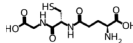
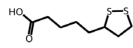
3.4. Classification of the Known Prooxidants and Antioxidants. The groups of prooxidants and antioxidants (Table S2) were distinguished by increased and decreased oxidative stress marker levels, respectively. ClassyFire³⁸ was used to provide an overview of the structural composition of these chemical modulators (Figure S8). Prooxidants were composed of 1284 organic compounds and 132 inorganic compounds, while antioxidants were made up of 1081 organic compounds and few inorganic compounds ($n = 21$). Both groups shared the same major superclasses, including benzenoids (24% of the prooxidants and 14% of the antioxidants), organoheterocyclic compounds (18% and 18%), organic acids and derivatives (12% and 10%), lipids and lipid-like molecules (10% and 20%), and organic oxygen compounds (7% and 6%). Notably, phenylpropanoids and polyketides were a major superclass uniquely for antioxidants

(20%), with most substances being plant antioxidants such as cinnamic acids, tannins, coumarins, flavonoids, and isoflavonoids.

As shown in Figure 4, chemical categories were further annotated by comparing with chemical information from the CompTox Chemicals Dashboard²⁹ and other publicly available databases (Table S3), focusing on the sources, functions, uses, and structures of significant attention.³⁹ In total, 81% of the prooxidants and 71% of the antioxidants were covered in the databases. The leading chemical categories were found as pharmaceuticals (20%), pesticides (12%), and metals and inorganic metal salts (6%) for prooxidants and pharmaceuticals (24%) and flavonoids (11%) for antioxidants. These findings align with the generic summaries on the main types of antioxidants and prooxidants in review articles.^{40,41} Several well-known chemical categories were predominantly prooxidants, including polycyclic aromatic compounds, polybrominated diphenyl ethers, polychlorinated biphenyls, per- and polyfluoroalkyl substances, bisphenols, algal toxins, fungal toxins, disinfection byproducts, flame retardants, and plastic additives. These chemicals accounted for 19% of the prooxidants and 3% of the antioxidants. Rather, certain phytochemicals (i.e., cinnamic acids and derivatives, tannins, and plant polyphenols) were mostly antioxidants, representing 7% of the antioxidants and only 1% of the prooxidants.

3.5. Characterization of the Most Concerning Prooxidants and Antioxidants. The top 10 prooxidants ranked by the number of records included several well-known pollutants (i.e., cadmium, arsenic, lead, chromium, copper, benzo[*a*]pyrene, and bisphenol A, Table 1), which are not only prevalent in the environment but also highly toxic, exhibiting carcinogenicity, neurotoxicity, and endocrine disruption. Epidemiological evidence links these pollutants to various

Table 1. Top Compounds Ranked by the Number of Records, Including the Top 10 Prooxidants and the Top 10 Antioxidants^a

Prooxidants				Antioxidants			
Rank	Name	CAS RN	Structure	Rank	Name	CAS RN	Structure
1	Cadmium	7440-43-9	-	3	Curcumin	458-37-7	
2	Inorganic arsenic	7440-38-2	-	4	Quercetin	117-39-5	
6	Doxorubicin	23214-92-8		5	Vitamin E	59-02-9	
8	Ethanol	64-17-5		7	Resveratrol	501-36-0	
9	Benzo[a]pyren	50-32-8		11	Melatonin	73-31-4	
10	Lead	7439-92-1	-	18	Inorganic selenium	7782-49-2	-
12	Bisphenol A	80-05-7		19	Trolox	53188-07-1	
13	Cisplatin	15663-27-1	$\text{Cl}-\text{Pt}^{2+}-\text{Cl}$ NH_3NH_3	20	Epigallocatechin gallate	989-51-5	
14	Chromium	7440-47-3	-	27	Glutathione	70-18-8	
15	Copper	7440-50-8	-	29	Thioctic acid	1077-28-7	

^aFor metals, metalloids, and selenium, the records for elements and their inorganic compounds were combined for ranking, and the CAS RN provided in the table corresponds to the element; for instance, the records for cadmium, cadmium chloride, and cadmium sulfate were merged.

diseases, including cardiovascular diseases, diabetes, and liver and kidney disorders.^{42–45} Oxidative stress plays a crucial role in these adverse outcomes through mechanisms such as DNA damage, chronic inflammation, and apoptosis.^{42–45} It stands to reason that understanding the mechanisms of oxidative damage caused by these pollutants and finding therapeutic agents to mitigate their toxicity have become a significant research concern.

The top 10 prooxidants also included pharmaceuticals (i.e., cisplatin and doxorubicin) and alcohol. Cisplatin and doxorubicin are widely used as chemotherapeutic agents with severe oxidative damage-related side effects, specifically nephrotoxicity,⁴⁶ ototoxicity,⁴⁷ and cardiotoxicity.⁴⁸ Therefore, intensive efforts are dedicated to finding protective agents, such as dietary antioxidants, free radical scavengers, thiol-containing compounds, and iron chelators, to enhance drug safety and reduce limitations on patient eligibility and

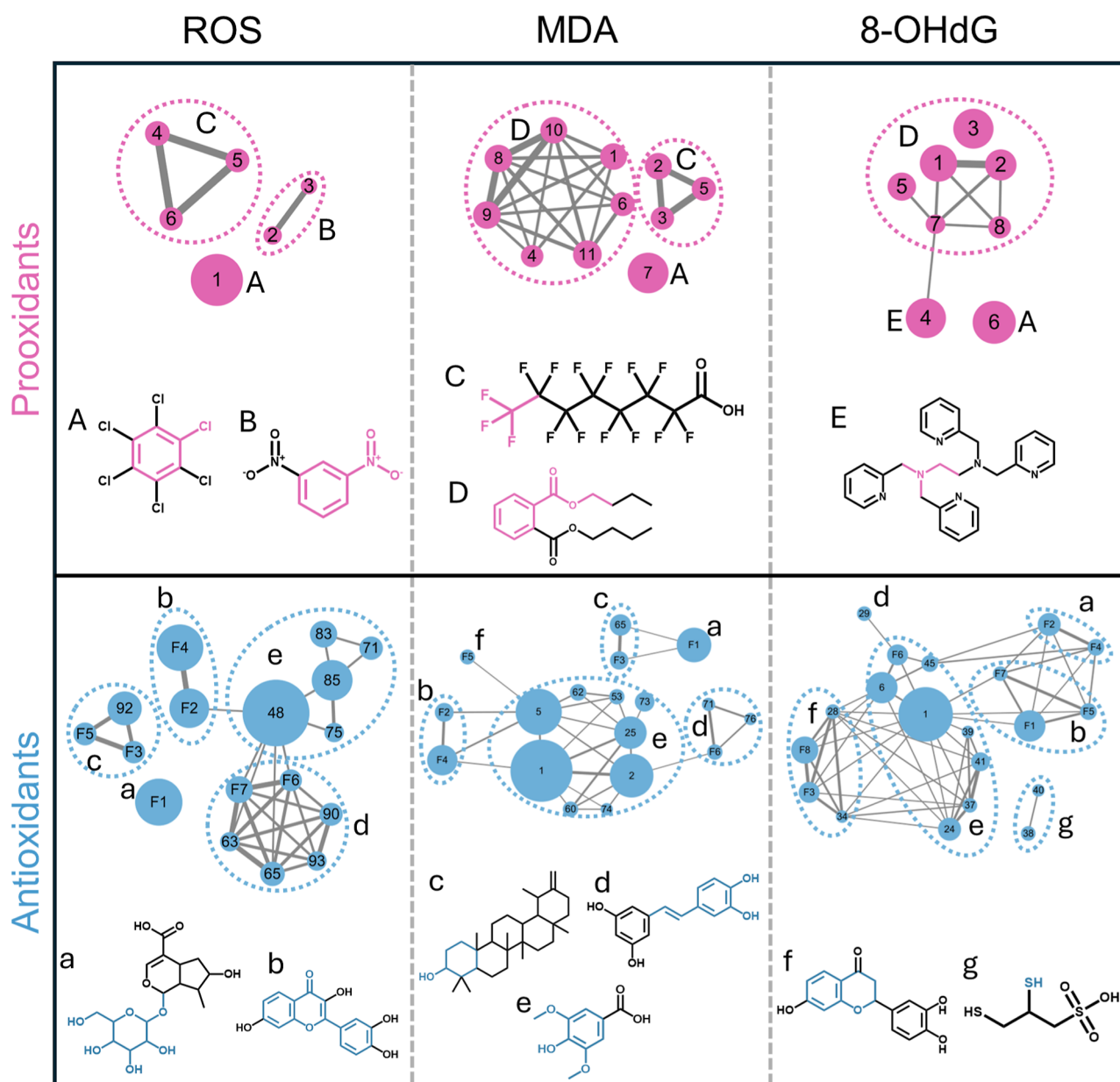


Figure 5. Network analysis grouped the prooxidant SAs into five types and the antioxidant SAs into seven types. Thickness of the line indicates the co-occurrence frequency, with thicker lines representing higher frequencies. Similarly, the size of the node reflects the frequency of the SA itself, with larger nodes indicating higher frequencies. Numbers in the nodes represent the identifiers of SAs, and all SAs are shown in Table S4. The capital letters A–E represent the types of prooxidant SAs: (A) chlorobenzene, (B) nitrobenzene, (C) fluoroalkyl, and (D) oxygen-containing groups, including alcohols, ethers, aldehydes, ketones, acids, and esters, and (E) tertiary amines. The lowercase letters a–g represent the types of antioxidant SAs: (a) hydroxyl, ether, and hydroxymethyl-substituted tetrahydropyran; (b) flavonoids; (c) hydroxylated cyclohexane; (d) alkenes, including meta/para-vinyl-substituted phenol, acrolein, acrylic, and acrylate; (e) hydroxyl and methoxyl-substituted benzene; (f) chromane; and (g) sulfur-containing groups, including thiol and sulfinyl. The red or blue parts in molecular structures are illustrations of representative fragments of each type.

treatment duration.⁴⁹ Ethanol is of particular interest because chronic excessive consumption leads to alcoholic liver disease, a significant health issue influenced by lifestyle, with oxidative damage being a key toxicological mechanism.⁵⁰

The top 10 antioxidants of interest included inorganic selenium, plant antioxidants (i.e., curcumin, quercetin, resveratrol, epigallocatechin gallate, and vitamin E), endogenous antioxidants (i.e., melatonin, thiocetic acid, and glutathione), and a synthetic antioxidant (Trolox). These substances have excellent antioxidant, anti-inflammatory, and anticancer activities with promising applications in drug

development and therapeutic strategies and are therefore of great research value in biomedicine and public health. Also, Trolox is widely used as a reference chemical to evaluate equivalent antioxidant capacity.⁵¹

Several plant antioxidants, including curcumin, quercetin, resveratrol, and epigallocatechin gallate, were found to exhibit antioxidant activity at lower doses but prooxidant activity at higher doses. In the case of doxorubicin-induced renal MDA in rats, the administration of 10 mg/kg quercetin reduced the resulting oxidative stress to the control level. However, this therapeutic effect diminished at 50 mg/kg, and an increase in

MDA levels was observed at 100 mg/kg.⁵² The prooxidant effects of these substances are usually observed on cancer cells, and a potential explanation is that cancer cells are more sensitive to these antioxidants compared to normal cells, exhibiting prooxidant effects at much lower doses. It is therefore hypothesized that the optimal dose range for a protective action would be to show prooxidant effects on cancer cells and antioxidant effects on normal cells.⁵³

3.6. Identification of Structural Alerts and Supporting Evidence. SA analysis was conducted to identify oxidative stress-related substructures. A diverse range of SAs were recognized for ROS, MDA, and 8-OHdG, with 6, 11, and 8 fragments in prooxidants and 95, 76, and 56 fragments in antioxidants, respectively (Table S4). Some fragments occurred simultaneously in the same compound, especially in compounds with antioxidant activity. These represented the same or highly similar substructures but with different atomic numbers when counted repeatedly and were hereafter referred to as “fragment families”. Network analysis made it easier to identify fragment families and clarify the relationships between different fragments. For antioxidant SAs, 7, 6, and 8 families were found with a co-occurrence frequency of $\geq 80\%$ in the ROS-, MDA-, and 8-OHdG-related fragments.

Based on expert judgment with supporting evidence, five types of prooxidant SAs were grouped from the oxidative stress-related substructures, as shown in Figure 5. These SAs could act through several potential mechanisms: (1) Metabolic conversion to free radicals. For instance, chlorobenzene (SA type code: A) and nitrobenzene (B) can be metabolized in vivo to semiquinone radicals. Nitrobenzene (B) can also be transformed to nitroanion radicals.^{54–57} (2) Induction of mitochondrial toxicity. The fluoroalkyl (C) exhibits strong lipophilicity, which can alter membrane structure and function, leading to mitochondrial dysfunction and potentially forming a self-amplifying cycle with ROS generation.⁵⁸ (3) An electrophilic mechanism, where electrophiles consume thiol-containing antioxidants such as glutathione, thereby reducing antioxidant capacity and increasing oxidative stress. Quinones and epoxides generated from chlorobenzene (A) metabolism, as well as quinones and nitrenium ions produced from nitrobenzene (B) metabolism, are all electrophiles.^{54–57} Acyclic carbonyl groups (D) are considered as SAs for electrophiles.⁵⁹ The *N*-oxide metabolites of tertiary amines (E) are also electrophilic.⁶⁰ Studies have shown that representative compounds of each type, such as hexachlorobenzene (A),⁶¹ *m*-dinitrobenzene (B),⁶² dibutyl phthalate (D),⁶³ and trimethylamine *N*-oxide (E),⁶⁴ can reduce glutathione levels. These multifaceted mechanisms highlight the potential role of the identified substructures in contributing to oxidative stress, which could be crucial in assessing and managing the environmental impact of chemical pollutants.

Seven types of antioxidant SAs were also obtained and may operate through various mechanisms: (1) By scavenging of free radicals. Flavonoids (SA type code: b), hydroxyl- and methoxyl-substituted benzene (e), and chromanes (f) typically contain phenolic hydroxyl or methoxyl groups. The benzene ring can activate hydrogen or methyl through conjugation, making them easier to leave and thus form stable phenoxyl radicals, which can then capture and neutralize free radicals.^{65,66} Alkenes (d) can also transfer electrons through conjugation, thereby positively affecting the antioxidant properties of the molecules. Additionally, thiol (g) and hydroxyl (a, b, c, e, and f) groups can directly scavenge free

radicals by donating hydrogen or transferring electron.⁶⁷ (2) By activating the Nrf2/ARE pathway. It leads to upregulation of the expression of antioxidant enzymes and substances encoded by downstream genes. For instance, among all 459 compounds containing the identified antioxidant SAs, 65 were tested in the Nrf2/ARE reporter gene assay of Tox21, and 21 were proven to be agonists. These agonists covered the full seven types of SAs, including flavonoids (a, such as apigenin and dihydroquercetin) and hydroxyl- and methoxyl-substituted benzenes (e, such as curcumin and resveratrol). (3) By chelation of metal ions. Compounds containing hydroxyl (a, b, c, e, and f) and thiol (g) groups can easily chelate metal ions (e.g., Fe and Cu), which play catalytic roles in ROS generation, thereby reducing ROS production.^{65,66,68} (4) By consumption of electrophilic reagents. Thiol (g) groups can react with electrophilic reagents (such as peroxynitrite and hypochlorous acid) to form adducts.⁶⁹ These pieces of evidence provide a mechanistic explanation for the modulating activity of each type of identified SAs, strengthening the rationality and reliability. In the meantime, there is still a paucity of knowledge on the interaction between different SAs or between one SA and the other substructures, and thus, more comprehensive toxicological studies are needed.

4. ENVIRONMENTAL IMPLICATIONS

In this study, an integrated workflow utilizing GPT-4 was developed to extract information on oxidative stress tests, consisting of data collection, text preprocessing, prompt engineering, and performance evaluation procedures. This workflow was an initial attempt to apply advanced LLMs to complex text mining tasks with multiple entities in the field of environmental sciences. Extensive records of oxidative stress tests ($n = 17,780$) were obtained, which led to in-depth insights into the trends of oxidative stress-related research such as the recent surge in research interest. In addition, a comprehensive literature-sourced list of prooxidants ($n = 1416$) and antioxidants ($n = 1102$) was established, focusing on chemicals in the environment. Based on this list, various types of SAs were identified, which enhances the understanding of the relationship between molecular structures and oxidative stress-modulating activities.

The obtained results demonstrated reliability under real-life (i.e., indoor dust, atmospheric fine particulate matter, and electronic waste) exposure scenarios. A total of 46 environmental chemicals associated with oxidative stress markers in human samples (e.g., urine and blood) were manually curated from the literature, including metals and metalloids, phthalates, organophosphate flame retardants and plasticizers, polycyclic aromatic hydrocarbons, parabens, bisphenols, volatile organic compounds, and antimicrobials (Appendix S6 and Table S5). Almost all chemicals were captured by the text mining results except for two metals (i.e., barium and strontium), resulting in a coverage rate of 96%. Applicability of the text mining results was further explored by matching the compiled list of chemical modulators and SAs with a database of indoor dust chemicals ($n = 511$),⁷⁰ where a significant number of known prooxidants ($n = 137$) and potential prooxidants ($n = 119$) were noticed. Among these known prooxidants, 52 were found to contain at least one of the five types of prooxidant SAs. It strengthened the evidence for oxidative stress as a critical mechanism of indoor dust-induced toxicity (Table S5).⁷¹ The chemical category with the highest number of compounds was pesticide, which consisted of 30 known and 18 potential prooxidants. It

was followed by flame retardants, volatile organic compounds, plasticizers, and polychlorinated biphenyls (Figure S9). These chemicals can serve as a suspect list for actual indoor dust samples, and further correlation analysis with data on oxidative stress-related diseases may have the potential to prioritize noteworthy hazardous chemicals. The successful establishment of the oxidative stress database illustrates the feasibility of building toxicological databases in a cost-efficient manner through LLM-based text mining.

The established workflow performed well in extracting information related to oxidative stress tests, providing a high-quality template (particularly in text preprocessing and prompt engineering) that can be conveniently adapted to other toxicological end points. It is expected to evolve into a general method for extracting toxicological information. For such generalized tasks, beyond prompt engineering, fine-tuning or a combination of both techniques should be considered when resource investment is justified by the scale of the task. Additionally, it is important to note that the current workflow handles only textual information, and future improvements could incorporate automatic recognition and parsing technologies for graphs and tables to further enhance the comprehensiveness of information extraction.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.4c07390>.

Search query; prompt; size calculation for validation set; SA analysis procedure; research trends; data collection for real-life exposure scenarios; characteristics of the workflow; compiled chemical category lists; SA analysis procedure; word frequency change; temporal changes of various markers, results, subjects, durations, and chemicals; structural classification by ClassyFire; and prooxidants in indoor dust (PDF)

Literature records and chemicals obtained from text mining; SAs; environmental chemicals in exposure scenarios; and chemical modulators in indoor dust (XLSX)

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Notes

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